area as bad sectors.

If you don't run a disk utility such as Scandisk or Norton Disk Doctor, the operating system continues to think of this area as a legible sector, and tries to read/write data to it. Failing to do so, it gives error messages indicating that the disk is inaccessible. You might also experience random lockups or crashes while using such a disk.

What's recoverable, what's not

The good news is that everything is recoverable—everything! Even if your data is inaccessible due to corrupt or demagnetised media. But at what price and how?

Physical errors or electrical failures require the disk to be opened and the components replaced, and hence require a qualified technician at a data recovery centre to perform these operations.

If your hard disk has been physically damaged, then recovery software would not be of any help. At the data recovery centre, the drive undergoes various tests and in the case of a drive electronics error, that is, your logic card being non-functional, it can simply be replaced.

A read error indicates that the drive motor has failed, or the head has been damaged, or it may not be aligned properly. In case of a media error the platters would have to be physically removed from the spindle and placed on a separate drive mechanism. They will then be read from there through custom-made recovery software, which works on the principle of trial and error, and will try to recover the data.

However, sending your hard disk to a data recovery centre involves a lot of expenses, as the charges levied are not based on the amount of data to be recovered or the amount of data that could be recovered. Instead, the charges are based on the importance of the data to the client and you could end up paying a bill worth a couple of lakhs of rupees. In India, such services are provided by Stellar Phoenix (***www.stellarinfo.com***).

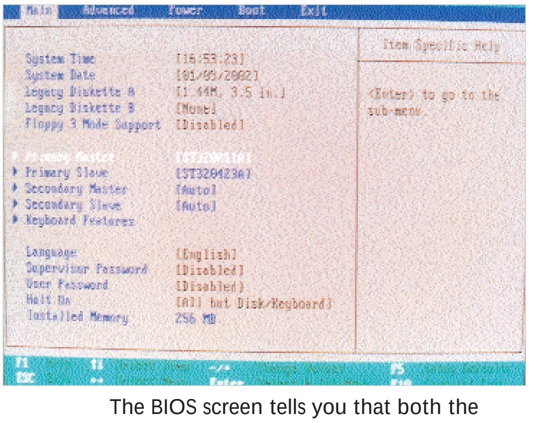
Average life of a hard disk should be around three to five years, considering an average practical usage of six to eight hours every day

If your hard disk is not physically damaged, you can do a simple file recovery at home using custom disk tools/recovery software such as Ontrack Easy Recovery and Lost & Found. Remember though, that this can be done only if your drive is detected by your BIOS. You would need to connect another hard drive to the machine to recover data to; you can't use the same hard drive for recovering files.

On the road to recovery

Here's how you can go about recovering your data:

STEP 1 As soon as you realise your hard disk has crashed, first shut down your machine so as to prevent any further damage to your disk. Then restart your machine, go to the BIOS and check if the hard disk is being detected. If your hard disk is not detected, you will have to send it to the data recovery centre. However, if the hard disk is detected then there are chances you can recover your data through a recovery software.



The BIOS screen tells you that both the hard disks have been detected

STEP 2 Next, plug in another hard disk to your computer so that the data you recover can be stored on it. However, ensure that the hard disk has more free space than the size of the data you need restored, else the task won't be successful. If you wish, you can plug in a zip drive also, but then make sure that the recovery software you're using supports such media. Remember that even though you might have another partition on the crashed hard disk, you can't recover data

The Root of the Trouble

The most common causes of damage to hard disks are mishandling and bad power. You can extend the life of your hard disk by:

- Not plugging in many drives to the same power cable
- Making sure there is no voltage fluctuation, or else the logic card could be damaged, thereby rendering the drive unusable.
- Taking great care while installing or removing the hard disk from your computer—a small jerk or bump could be enough to damage your drive, sometimes.

Keeping Your Disk Fighting Fit

While some hardcore geeks may just never be content with their PC's speed, most of us are genuinely plagued by slow computers. Of the several factors that could lead to a slow performing system, the hard disk is an important one.

In order to get the top-notch performance you need from your hard disk you should, as a thumb rule, run the Scandisk program bundled with Windows which scans the hard disk for surface errors. The hard disk consists of the file allocation table (FAT), which indexes all the entries of files. In the case of an entry being damaged, your files may be lost or, at worse, your system may not boot up. Hence, by periodically running Scandisk you automatically eliminate severe damage to files that may be caused during normal use of the system or even due to an improper shutdown of your machine.

'Disk Defragmenter' is another utility that comes bundled with Windows. This utility arranges all the files systematically so that the hard disk does not have to waste time hunting for the rest of the program files while trying to open a particular program. Hence, if you frequently install and uninstall programs, thereby causing changes in the file arrangement, then you should run this utility at least once a month so that your hard disk performance does not get affected.